THE ZERO PARADOX

ZP-P: The Fixed-Point Fork

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Synthesis layer. The abstract fork schema is proved sorry-free and choice-free in Lean 4 (FixedPointFork.lean); the number-system instance via Ostrowski (Ostrowski.lean) and the categorical-parent instance via QPF.Fix / QPF.Cofix (Coalgebra.lean) are Lean-witnessed; the set-theory and computation instances are referenced to ZP-J and ZP-K. Generalizes the ZFC+Foundation / ZFC+AFA orthogonal-contact-point claim.

A self-referential operator over a complete lattice has a least fixed point and a greatest fixed point (Knaster–Tarski). ZP-P records a single elementary fact and develops its consequences across the framework: these two fixed points coincide — the fork collapses to one point — exactly when the operator has a unique fixed point. Read intuitively, the least fixed point is the inductive (well-founded) closure and the greatest is the coinductive (non-well-founded) closure; the collapse point is where the two closures meet. In the Zero Paradox that point is the diagonal fixed point ⊥, the self-containing bottom.

This layer is a synthesis layer, not a foundational one: it adds no axiom or primitive. It names a pattern that several existing layers already realise, and it is organised in three tiers held to three different standards of proof. Tier 1 (Section I) is the abstract schema, proved in Lean over a complete lattice. Tier 2 (Section II) is the instance catalogue: each framework forks at its own contact point, and the instances are theorems in their home layers. Tier 3 (Section III) is the unification — that the contact points are faces of one object — which is a fenced conjecture, never a formal claim. The fences in Section III are load-bearing: keeping the tiers separate is what makes the formal content honest.

Section I: The Fork Schema (Tier 1)

Over a complete lattice, a monotone self-map f has a least fixed point (lfp f) and a greatest fixed point (gfp f), and lfp f ≤ gfp f always. The width between them is the fork. The schema theorem states that the fork collapses to a single point precisely when f has a unique fixed point — and in that case the common value is that fixed point. The proofs rest only on Mathlib's order-theoretic fixed-point API (OrderHom.lfp / OrderHom.gfp); they make no claim about the categorical inductive/coinductive reading, which is offered as analogy only.

Definition: the fork (FixedPointFork.lean)
For a complete lattice α and a monotone self-map $f : \alpha \to \alpha$:
lfp f = the least fixed point of f (Mathlib OrderHom.lfp)
gfp f = the greatest fixed point of f (Mathlib OrderHom.gfp)
The fork is the ordered pair (lfp f, gfp f), always satisfying lfp f ≤ gfp f.
Collapse = lfp f = gfp f: the two ends meet at a single contact point.

Proposition: fork_le (FixedPointFork.lean)

lfp f ≤ gfp f

The inductive closure never exceeds the coinductive closure: the fork has non-negative width. (Mathlib OrderHom.lfp_le_gfp.)

Lean purity: [propext, Quot.sound] — choice-free. ✓

Lemma: collapse_of_unique (FixedPointFork.lean)

If x is the unique fixed point of f, then lfp f = x and gfp f = x.

Both ends of the fork land on the sole fixed point.

Proof: lfp f and gfp f are fixed points (map_lfp, map_gfp); uniqueness sends each to x.

Lean purity: [propext, Quot.sound] — choice-free. ✓

Lemma: unique_of_collapse (FixedPointFork.lean)

If lfp f = gfp f, then every fixed point of f equals that common value.

Every fixed point lies between lfp f and gfp f, so collapse forces uniqueness.

Proof: lfp_le_fixed and le_gfp bracket any fixed point y; antisymmetry closes it.

Lean purity: [propext, Quot.sound] — choice-free. ✓

Theorem: fork_collapse_iff (FixedPointFork.lean) — the schema spine

lfp f = gfp f ↔ ∃! x, f x = x

The fork collapses to a single contact point if and only if f has a unique fixed point.

This is the abstract form of "the framework forks, and the diagonal fixed point is where the two closures meet."

Proof: collapse_of_unique and unique_of_collapse, both directions.

Lean purity: [propext, Quot.sound] — choice-free. ✓

The fork spine is choice-free. All four results depend only on [propext, Quot.sound] — propositional extensionality and quotient soundness, the benign kernel axioms used throughout Mathlib. None depends on the Axiom of Choice. This is consistent with the choice-free core (see AxiomProfile.lean): the conceptual skeleton needs no choice; choice enters only in the analytic realisations (Section II).

Section II: The Instance Catalogue (Tier 2)

The same fork appears across frameworks. ZP-P does not re-prove each instance — each is a theorem in its home layer — it records the catalogue and pins the number-system instance to a checkable Lean witness. The original ZFC+Foundation / ZFC+AFA orthogonal-contact-point claim is the set-theory entry: in the standard category-theoretic characterisation (Aczel; Lambek), Foundation is the initial algebra (μ) and AFA the final coalgebra (ν) of the bounded powerset functor, and the Quine atom $\perp = \{\perp\}$ is exactly an element present in ν but not in μ.

Domain	The two closures fork into	Contact point / home layer
Set theory	Foundation (μ, well-founded) vs AFA (ν, non-well-founded)	Quine atom $\perp = \{\perp\}$; ZP-J / ZP-K
Number systems	Archimedean $\mathbb{R}$ vs non-Archimedean $\mathbb{Q}_p$	0 (snap fails in $\mathbb{R}$, holds in $\mathbb{Q}_2$); ZP-B / ZP-F
Computation	total (well-founded) vs partial (non-terminating)	the Kleene quine / self-application; ZP-K
Category theory	initial algebra (W-types) vs final coalgebra (M-types)	QPF.Fix vs QPF.Cofix; Coalgebra.lean

The number-system instance is the one with a genuine classification theorem behind it. Ostrowski's theorem says the nontrivial absolute values on $\mathbb{Q}$ are exactly the real (Archimedean) one and the p-adic (non-Archimedean) ones — the complete, pairwise-inequivalent list. The contact point is 0: in the Archimedean completion $\mathbb{R}$ it is approached but never reached (density — the snap fails), while in the 2-adic completion $\mathbb{Q}_2$ it is the unique fixed point of $x \mapsto 2x$ and its valuation diverges ($v_2(0) = \infty$ — the snap holds).

Theorem: completions_exhaustive (Ostrowski.lean)

Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to the real absolute value, or to a p-adic absolute value for a unique prime p.

The two kinds of completion are the complete list. (Ostrowski's theorem, Mathlib Rat.AbsoluteValue.equiv_real_or_padic.)

Lean purity: [propext, Classical.choice, Quot.sound] — choice-carrying. ✓

Theorem: real_not_equiv_padic (Ostrowski.lean)

The real absolute value is inequivalent to every p-adic absolute value.

The two kinds of completion are genuinely distinct — never the same metric. This is the orthogonality leg of the number-system fork. (Mathlib Rat.AbsoluteValue.not_real_isEquiv_padic.)

Lean purity: [propext, Classical.choice, Quot.sound] — choice-carrying. ✓

Remark: choice-free spine, choice-carrying realisation

The fork spine (Section I) is choice-free [propext, Quot.sound]. The Ostrowski instance carries Classical.choice, inherited from Mathlib's classical analysis and number theory. This split is the framework's standard pattern: the conceptual core is choice-free, while the analytic realisations are choice-carrying.

Remark: choice-free spine, choice-carrying realisation

The number-system fork is theorem-backed on its own terms — Ostrowski is a genuine classification theorem, stronger backing than the metatheoretic Foundation/AFA orthogonality. It is NOT claimed to be an instance of the μ/ν (least-vs-greatest fixed point) schema: Ostrowski concerns absolute values, not fixed points of a functor. The thread to the diagonal fixed point runs through the contact point 0 (the unique fixed point of $x \mapsto 2x$ in $\mathbb{Q}_2$), not through the schema theorem.

The categorical instance is the genuine μ/ν case — the parent of the set-theory entry. On a one-shape, one-child polynomial functor (no leaf), the initial algebra (W-type, μ) is empty while the final coalgebra (M-type, ν) is inhabited by the infinite self-referential element: the non-well-founded closure contains a point the well-founded closure lacks — the categorical analog of the Quine atom present in νF but not μF.

Theorem: categorical_fork_strict (Coalgebra.lean)

IsEmpty (Fix idPF_Coalgebra.Obj) ∧ Nonempty (Cofix idPF_Coalgebra.Obj)

The initial algebra (least fixed point, μ) is empty; the final coalgebra (greatest fixed point, ν) is inhabited. (Mathlib QPF.Fix / QPF.Cofix.)

Split footprint: fix_isEmpty (μ empty) is choice-free [propext, Quot.sound]; cofix_nonempty (ν inhabited) carries Classical.choice from the M-type / corecursion machinery — a library artifact, not a necessity: for a polynomial functor the final coalgebra is constructible choice-free (Ahrens–Capriotti–Spadotti, TLCA 2015). ✓

Remark: where choice genuinely enters the μ/ν fork

The Classical.choice in cofix_nonempty is inherited from Mathlib's M-type machinery, not forced by the mathematics: for a polynomial functor the final coalgebra is constructible choice-free (Ahrens–Capriotti–Spadotti, TLCA 2015, who construct it as an ω-limit). Their ambient setting is intensional type theory with univalence, but the construction itself uses only function extensionality — univalence enters only for their uniqueness result.

Veltri (FSCD 2021) points the other way and is cited here for contrast, not support: his subject is the finite-powerset functor, which is not polynomial, and there the choice-free route is the one that is hard — his finality results are obtained assuming choice principles rather than shown to need them, and the one genuine necessity he proves is of LLPO, a logic taboo, not of choice. He cites Ahrens–Capriotti–Spadotti for the polynomial case himself.

Choice genuinely enters only for the non-polynomial finite-powerset functor, where the literature pins each presentation: the set-quotient's finality proof assumes the full axiom of choice, while Worrell's $(\omega+\omega)$-limit is carried out under countable choice together with the lesser limited principle of omniscience (LLPO). Injectivity of the canonical algebra implies LLPO outright, and is equivalent to it under countable choice (Veltri, FSCD 2021).

The remaining two instances are referenced, not re-proved here. The set-theory fork (Foundation vs AFA) has its Lean witness in ZP-J: the Quine atom $\bot = \{\bot\}$ is the unique self-application fixed point (quine_atom_unique). The computation fork (total vs partial) has its Lean witness in ZP-K: the Kleene quine. Each is the contact point of its own fork.

Section III: The Unification and Its Fences (Tier 3)

The tempting claim is that the contact points of all the instances — the Quine atom, the 2-adic 0, the Kleene quine, the categorical initial object — are faces of one object, the diagonal fixed point. ZP-P states this as a conjecture and fences it precisely. The fences are not hedging; they mark a genuine boundary, and per-instance forks remain full theorems on either side of them.

Hard fence (permanent): cross-instance identity is not a theorem

The claim "the contact points are one object" is not a theorem and cannot become one. Identity requires a shared type: the 2-adic 0, the Quine atom (a set), the Kleene quine (a code), and a categorical initial object (an object up to isomorphism) are terms of different types in different categories. The proposition "x = y" across them is not false — it is not well-formed. So this is a type boundary, not a missing proof. What is claimed formally is only that each framework forks at its own contact point.

Soft fence (conjecture): not every fork is a μ/ν instance

That every domain fork is an instance of the least-vs-greatest fixed point schema is an open generalisation, not established here. The schema is a theorem in this layer and for canonical functors (W-types vs M-types). The number-system fork (Ostrowski) is the clear case that is theorem-backed yet NOT visibly a μ/ν instance — it is the live caution against overstating the generalisation.

Per-instance forks are theorems and are not hedged. The fences scope only the cross-instance identity (Tier 3) and the universal schema claim. Each individual fork — Foundation/AFA, $\mathbb{R}/\mathbb{Q}_2$, total/partial — stands on its own proof.

Theorem Summary

Result	File / Section	Axioms
fork_le	FixedPointFork.lean §I	choice-free
collapse_of_unique	FixedPointFork.lean §I	choice-free
unique_of_collapse	FixedPointFork.lean §I	choice-free
fork_collapse_iff	FixedPointFork.lean §I	choice-free
completions_exhaustive	Ostrowski.lean §II	Classical.choice
real_not_equiv_padic	Ostrowski.lean §II	Classical.choice
fix_isEmpty	Coalgebra.lean §II	choice-free
cofix_nonempty	Coalgebra.lean §II	Classical.choice

Axiom Purity

Fork spine (FixedPointFork.lean): [propext, Quot.sound] — choice-free. The schema needs no Axiom of Choice.

Number-system instance (Ostrowski.lean): [propext, Classical.choice, Quot.sound] — Classical.choice inherited from Mathlib's classical analysis / number theory (Ostrowski).

Axiom Purity
Categorical instance (Coalgebra.lean): split — fix_isEmpty (μ empty) is choice-free [propext, Quot.sound]; cofix_nonempty (ν inhabited) carries Classical.choice from the M-type / corecursion machinery.
Core choice-free, realisation choice-carrying — the framework's standing pattern. Zero sorry. Verified: lake build, August 2026.

End of ZP-P | The Fixed-Point Fork | fork_collapse_iff (choice-free spine) | Ostrowski number-system instance | categorical μ/ν instance (Fix empty / Cofix inhabited) | hard fence: cross-instance identity is a type boundary | soft fence: not every fork is μ/ν | All FixedPointFork.lean / Ostrowski.lean / Coalgebra.lean theorems verified.